Closed-Loop Dynamic Thermal Management (DTM) for 3D Stacked HBM2 Architectures

A Comprehensive Engineering and Implementation Journey

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Introduction & Motivation

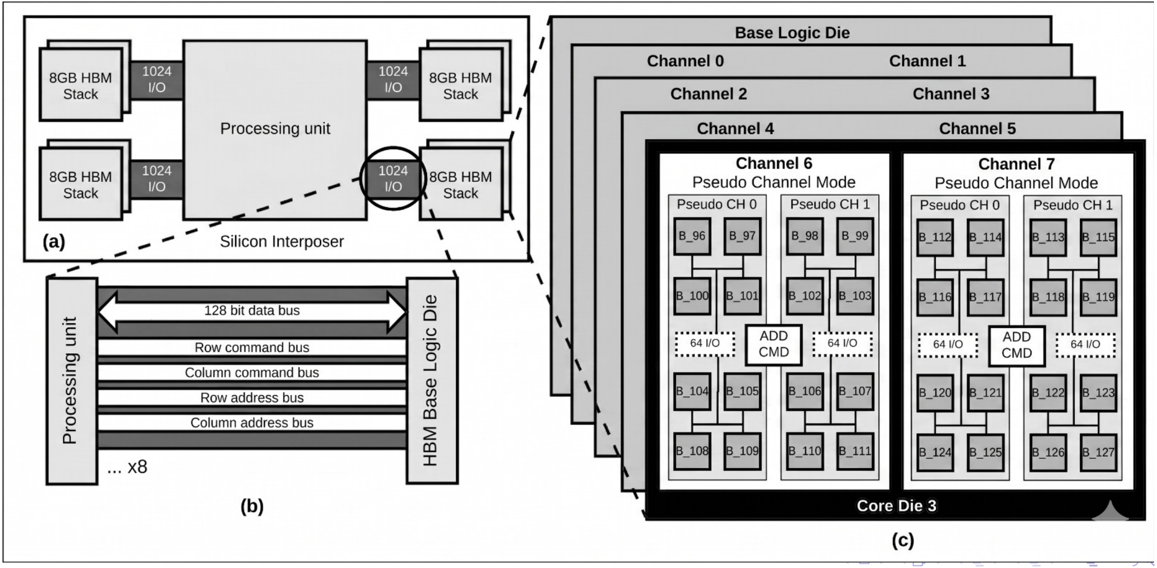
The Challenge with HBM2:

- Transition from 2D (GDDR6) to 3D stacked High Bandwidth Memory (HBM) revolutionized GPU bandwidth.
- **Thermal Constraints:** Stacking active silicon dies vertically introduces severe heating.
- High temperatures (e.g., 85°C) destroy data retention in memory capacitors.

Project Goal:

- Simulate thermal dissipation and real-time throttling of a 4 HBM2 stack under intense workloads.
- Create an accurate Dynamic Thermal Management (DTM) policy scheduler based on generated power traces.

HBM memory architecture



The Simulation Pipeline

The project was broken into four distinct phases to handle the multi-domain complexity:

1. **Microarchitectural Simulation:** Generate realistic DRAM access patterns.
2. **Power Modeling:** Translate raw DRAM commands into physical wattage.
3. **3D Thermal Modeling:** Build a physical HBM stack representation to calculate spatial heat flow.
4. **Closed-Loop Control:** Architect a software DTM controller to monitor and recursively throttle the simulation.

Phase I: Microarchitectural Trace Generation

Tooling & Setup:

- Used Accel-Sim / GPGPU-Sim with NVIDIA's Volta V100 (`QV100-SASS`) architecture.
- **Workload:** `SRAD` (Speckle Reducing Anisotropic Diffusion) from Rodinia 3.1.
- **Run specifics:** `sim_run_12.8/srad_v1-rodinia-3.1/100_0_5_502_458/QV100-SASS`
- Chosen due to heavy spatial memory access patterns, yielding a lot of activity to record and triggering DTM thresholds.

Rectifying HBM2 vs GDDR6:

- Expanded bus width to 16 bytes per channel.
- Reduced Burst Length (BL) to 2.
- Locked core clock to 850 MHz (256 bits per command).
- 8 partitions per stack, 16 banks per partition $\rightarrow$ 128 memory banks.

Phase II: Bridging Architecture to Power

The Power Formula:

$$P_{bank} = \frac{(C_{act} E_{act}) + (C_{pre} E_{pre}) + (C_{rd} \cdot B \cdot E_{rd}) + (C_{wr} \cdot B \cdot E_{wr})}{\Delta T}$$

C = Command count, $B = 256$ bits/command, $\Delta T \approx 1.176$ ns/cycle

The "Melt the Chip" Energy Bug:

- Initial thermal simulations hit 440 K (169°C).
- Problem identified as we used materials files from example which wasn't needed.

Phase III: 3D Thermal Modeling (HotSpot 3D)

Layer Configuration (LCF) for 4 Stack:

- 9 total vertical layers configured.
- **4 Active Silicon Dies** (power dissipation enabled).
- **5 Inactive TIM/Glue Layers** (power dissipation disabled).

Data Integration Quirks:

- TIM layers were completely ignored by HotSpot's parser. Python generator adjusted to output exactly 128 columns (32 banks $\times$ 4 layers).
- **Naming Convention:** Used unique bank names (`B_0` to `B_127`) to bypass HotSpot's layer prefix duplication logic.

Phase IV: Data Engineering & Optimization

The Bottleneck:

- Seconds of logical execution inflated CSV trace data to **30 GB** - **100 GB**.
- Standard Python loops failed under RAM constraints (16GB System RAM, 4GB VRAM).

Out-of-Core Streaming with Polars:

- Replaced standard `for`-loops with **Polars** (Rust-based DataFrame library).
- Used `LazyFrames` and streaming execution to parse data *out-of-core* directly from SSD.
- Processed 30 GB CSVs into 8 independent partition files without loading into system RAM.
- **Result:** Processing time plummeted from hours to **under 3 minutes**.

Phase V: Closed-Loop DTM Controller

The DTM Policy:

- **Throttle ON:** Breaching 355.00 K (82°C).
- **Throttle OFF:** Cooling below 353.00 K (80°C).

The $O(N^2)$ Trap vs. $O(1)$ "Stepper Motor":

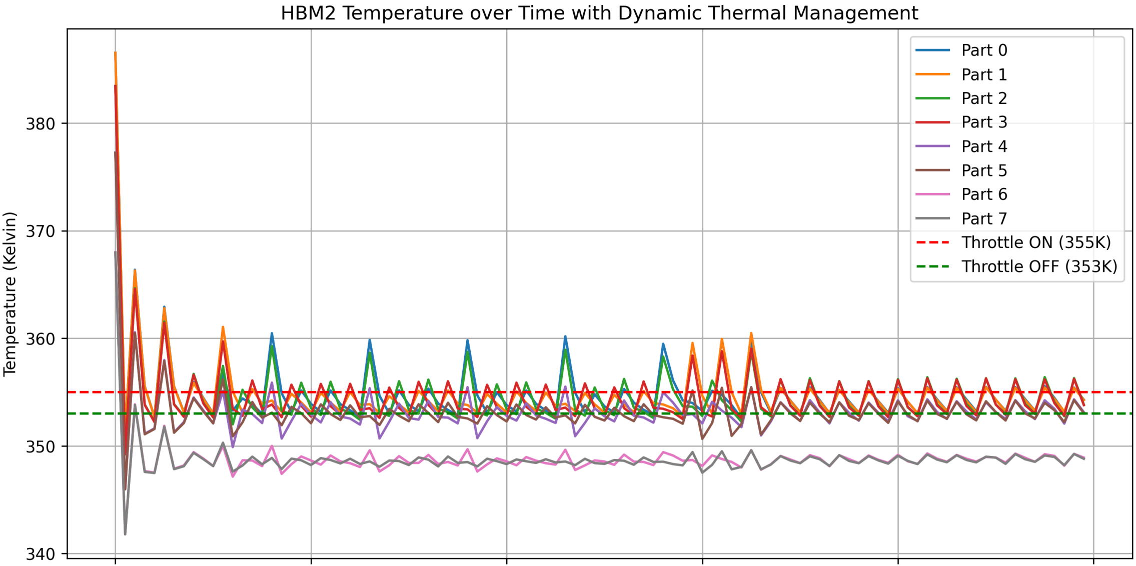
- Growing trace files recursively forced HotSpot into $O(N^2)$ recalculations.
- **Solution:** Developed a stateful "Stepper Motor". Passes exactly **1 row** of power data with an `.init` file representing previous thermal state.
- Loop execution time slashed to fractions of a second.

Phase V: Asynchronous Throttling Execution

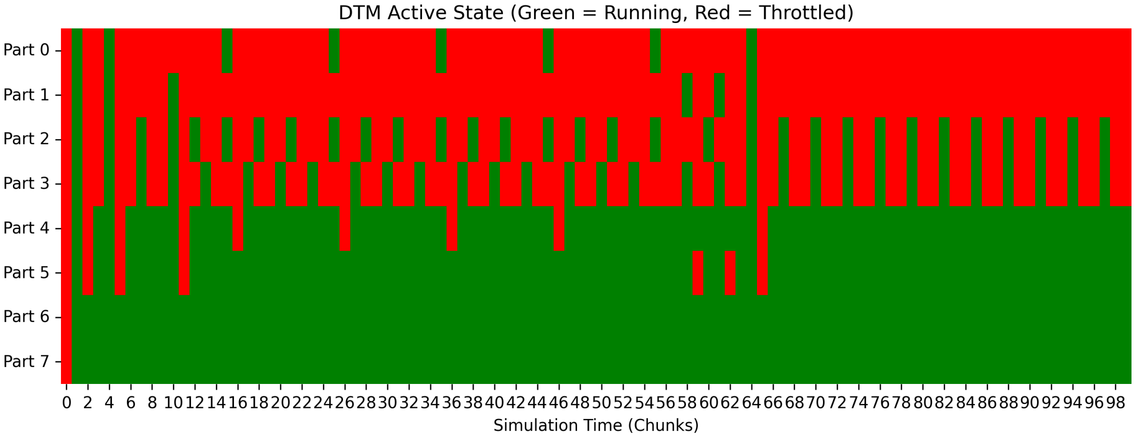
Asynchronous Partition Pointers:

- Handled throttling without desynchronizing memory trace data.
- Maintained 8 independent file handles with an `active_state` array.
- If Partition P_i is throttled, inject 0.0W array for P_i and leave its file pointer stationary.
- Perfectly mimics a stalled memory controller clock.

DTM Results


HBM2 Temperature over Time with Dynamic Thermal Management
Temperature (Kelvin)
380
370
360
350
340
Part 0
Part 1
Part 2
Part 3
Part 4
Part 5
Part 6
Part 7
Throttle ON (355K)
Throttle OFF (353K)

Actual trace file



Conclusion and Outcomes

- Built a robust, scalable, software-defined 3D thermal simulation framework for HBM2 architectures.
- Successfully generated high-activity traces from the `SRAD` benchmark on `QV100-SASS`.
- Solved massive data bottlenecks using a multithreaded **Polars** pre-aggregator.
- Engineered an optimized $O(1)$ Python stepper loop to interface with HotSpot 3D physics.
- The project lays a critical foundation for analyzing physical performance degradation and designing thermal-aware memory controllers.

Thank You

Questions?